JavaScript is disabled on your browser.

Java SE 27 & JDK 27

Class XMLFormatter

java.lang.Object
    java.util.logging.Formatter
        java.util.logging.XMLFormatter

```
public class XMLFormatter
extends Formatter
```

Format a LogRecord into a standard XML format.

The DTD specification is provided as Appendix A to the Java Logging APIs specification.

The XMLFormatter can be used with arbitrary character encodings, but it is recommended that it normally be used with UTF-8. The character encoding can be set on the output Handler.

Implementation Requirements:

Since JDK 9, instances of LogRecord contain an `Instant` which can have nanoseconds below the millisecond resolution. The DTD specification has been updated to allow for an optional `<nanos>` element. By default, the XMLFormatter will compute the nanosecond adjustment below the millisecond resolution (using `LogRecord.getInstant().getNano() % 1000_000`) - and if this is not 0, this adjustment value will be printed in the new `<nanos>` element. The event instant can then be reconstructed using `Instant.ofEpochSecond(millis/1000L, (millis % 1000L) * 1000_000L + nanos)` where `millis` and `nanos` represent the numbers serialized in the `<millis>` and `<nanos>` elements, respectively.
The `<date>` element will now contain the whole instant as formatted by the `DateTimeFormatter.ISO_INSTANT` formatter.

For compatibility with old parsers, XMLFormatters can be configured to revert to the old format by specifying a `<xml-formatter-fully-qualified-class-name>.useInstant = false` property in the logging configuration. When `useInstant` is `false`, the old formatting will be preserved. When `useInstant` is `true` (the default), the `<nanos>` element will be printed and the `<date>` element will contain the formatted instant.

For instance, in order to configure plain instances of XMLFormatter to omit the new `<nano>` element, `java.util.logging.XMLFormatter.useInstant = false` can be specified in the logging configuration.

Since:

1.4

Constructor Summary

Constructors

Constructor	Description
`XMLFormatter()`	Creates a new instance of XMLFormatter.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
`String`	`format(LogRecord record)`	Format the given message to XML.
`String`	`getHead(Handler h)`	Return the header string for a set of XML formatted records.
`String`	`getTail(Handler h)`	Return the tail string for a set of XML formatted records.

Methods declared in class Formatter

`formatMessage`

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Details

XMLFormatter

```
public XMLFormatter()
```

Creates a new instance of XMLFormatter.

Implementation Requirements:

Since JDK 9, the XMLFormatter will print out the record event time as an Instant. This instant has the best resolution available on the system. The `<date>` element will contain the instant as formatted by the `DateTimeFormatter.ISO_INSTANT`. In addition, an optional `<nanos>` element containing a nanosecond adjustment will be printed if the instant contains some nanoseconds below the millisecond resolution.

This new behavior can be turned off, and the old formatting restored, by specifying a property in the logging configuration. If `LogManager.getLogManager().getProperty( this.getClass().getName() +".useInstant")` is `"false"` or `"0"`, the old formatting will be restored.

Method Details

format

```
public String format(LogRecord record)
```

Format the given message to XML.

This method can be overridden in a subclass. It is recommended to use the `Formatter.formatMessage(LogRecord)` convenience method to localize and format the message field.

Specified by:

`format` in class `Formatter`

Parameters:

`record` - the log record to be formatted.

Returns:

a formatted log record

getHead

```
public String getHead(Handler h)
```

Return the header string for a set of XML formatted records.

Overrides:

`getHead` in class `Formatter`

Parameters:

`h` - The target handler (can be null)

Returns:

a valid XML string

getTail

```
public String getTail(Handler h)
```

Return the tail string for a set of XML formatted records.

Overrides:

`getTail` in class `Formatter`

Parameters:

h - The target handler (can be null)

Returns:

a valid XML string

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
Java is a trademark or registered trademark of Oracle and/or its affiliates in the US and other countries.
Copyright © 1993, 2026, Oracle and/or its affiliates, 500 Oracle Parkway, Redwood Shores, CA 94065 USA.
All rights reserved. Use is subject to license terms and the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.